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Persistence and near persistence via trait evolution: Pathways to coexistence

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ABSTRACT

In community ecology, predictions on whether or not interacting species will coexist are often formed by considering invasion criteria: each species must be able to increase when rare in a resident community at ecological equilibrium. However, when considering trait evolution, for example of competing species, this concept must be refined to account for evolution potentially feeding back on population dynamics. We analyze a model of two competing species whose quantitative traits evolve under stabilizing selection, assuming fixed but potentially unequal intraspecific trait variances. Using persistence theory, we derive conditions under which both species persist for all initial trait values, and we introduce the concept of 'near persistence', in which coexistence holds for a specific, biologically relevant, subset of initial trait configurations. Our analysis reveals that equal trait variances between species, which is commonly assumed in previous work, yield highly specific outcomes and obscure a broader set of coexistence scenarios that emerge under asymmetric trait variation. We identify fifteen distinct qualitative dynamical regimes as a function of initial conditions and trait variances. Moreover, we show that invasion analysis must often be performed at multiple eco-evolutionary equilibria, as invader dynamics may converge to different trait values depending on the ecological and evolutionary context. We discuss the biological implications of these results and perspectives for future work.

1. Introduction

Understanding the origin and maintenance of species diversity is a central goal in ecology and evolutionary biology. In community ecology, a major focus is answering the question of how multiple species persist despite constraints imposed by finite resources and interspecific interactions (Vellend, 2016). Many approaches have been applied to answering this question, including local-stability analysis e.g., (Allesina and Tang, 2012; Dougoud et al., 2018; Grilli et al., 2017; May, 1972), models of spatial heterogeneity e.g., (Chesson, 2000a; Hanski and Ranta, 1983; MacArthur and Wilson, 1967; Nee and May, 1992), or invasibility criteria e.g., (Barabás et al., 2018; Chesson, 2000b, 2018).

In the invasibility criteria approach, 'coexistence' is determined by each species having a positive growth rate when rare, suggesting that all species are expected to persist even if at low densities (Chesson, 2000b). This approach underlies modern coexistence theory, which partitions these coexistence criteria in such a way that informs us on the mechanisms that enable species coexistence e.g., Barabás et al., 2018; Chesson, 2000b, 2018. Historically, modern coexistence theory has focused on pairwise species interactions e.g., Chesson and Warner, 1981; Chesson, 1982, or invasion criteria for a focal species against a homogenised stable community e.g., Li and Chesson, 2016, although the framework can be extended to more complex communities (Ranjan et al., 2024). Although the invasion criteria is a unifying framework for investigating mechanisms of coexistence in community ecology, a critical assumption is that evolutionary dynamics are ignored, with the implicit assumption that evolution occurs on a much slower time scale than ecological dynamics. However, the nature and magnitude of interspecific

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interactions depend on potentially evolving species traits. In the past decades, there has been increasing evidence that evolution can occur on timescales rapid enough to affect ecological processes and dynamics, and thus also species interactions (Brunner et al., 2017; Ellner et al., 2011; Farkas et al., 2013; Thompson, 1998; Zamorano et al., 2023). Evolution of species traits can therefore alter the degree to which a focal species affects the density of other species and of itself, which in turn can alter the selective pressure on the trait resulting in further evolution. This interaction between ecological and evolutionary dynamics is also referred to as an eco-evolutionary feedback (Hairston et al., 2005; Hendry, 2016). Consequently, the potential for coexistence within a community might change due to these eco-evolutionary dynamics.

Eco-evolutionary dynamics have been modelled extensively (see e.g., Barabás et al., 2022; Cropp and Norbury, 2021; Govaert et al., 2019; Lion, 2018; Patel et al., 2018; Weyerer et al., 2023), but integration with coexistence theory remains limited. Recent attempts to investigate how trait evolution impacts propensity for coexistence have begun. For example, Pastore et al. (2021) applied an eco-evolutionary model to investigate stabilising and equalising mechanisms of coexistence by modelling the evolution of mean trait values affecting competition. Ecological dynamics in their model apply a Lotka-Volterra approach in which intrinsic population growth and the strength of competition both depend on a focal trait (Pastore et al., 2021). Evolution is underpinned by a version of the breeder's equation in which the trait distribution is normal with a fixed variance. Pastore et al. (2021) demonstrated how trait evolution could affect coexistence and identified eco-evolutionary equilibria at which coexistence is predicted. In another study, Yamamichi et al. (2022) adapted Chesson's classical coexistence theory to an eco-evolutionary framework in two competing species. More specifically, they derived expressions for niche overlap and competitive differences assuming single species dynamics go to an equilibrium point and that evolution is rapid enough for traits to stabilise prior to invasion such that both resident and invader are adapted to competition with the resident.

One problem that arises when extending invasion criteria to integrate evolution is that the ecological dynamics of species densities must be coupled to evolutionary dynamics of traits, which results in a higher-dimensional system. In higher-dimensional systems, there are two challenges to determining whether species will coexist: First, determining whether a species can invade when rare in this context is a non-trivial task because it is not obvious a priori at what trait values we should evaluate invasibility. Second, it is not clear whether mutual invasibility is sufficient to imply species coexistence (e.g. in three-species rock-paper-scissors dynamics).

Hence, a more precise analysis of this issue is needed. Here, we do this using the simple two-species competition model analysed by Pastore et al. (2021) and using the framework of persistence theory (Patel and Schreiber, 2018; Schreiber et al., 2011; Garay, 1989), in which all species are ensured to persist given initially positive densities. This notion of coexistence indicates that all species will continue to coexist, despite large perturbations in the system (Schreiber, 2006). We first characterise the eco-evolutionary equilibrium of the resident species and show that, depending on the parameter values (in particular the trait variances of each species), the invader can evolve to different trait values when rare in this model. We then analyse the potential for coexistence to find precise conditions that ensure coexistence, in the sense of persistence, for any initial trait values. We then highlight that the usual notion of persistence may be too restrictive and not biologically meaningful. To address this, we define a modified form of coexistence, termed *nearly persistent*, which restricts to a subset of initial trait values, and we give precise conditions for this weaker notion of coexistence. Finally, we explore and categorize the possible eco-evolutionary dynamics numerically using the desolve package in R (Soetaert et al., 2010). All code is available on GitHub (Lyberger et al., 2025).

2. A two-species eco-evolutionary model

We consider a model of two co-evolving and competing species. Individuals within a species are characterised by an underlying trait z that influences their fecundities and competition with conspecifics and heterospecifics. We track changes in population abundance $n_i(t)$ and trait distribution $\phi_i(z,t)$ for species i. Note that species have potentially different trait distributions. As in Pastore et al. (2021), we assume that the trait distribution for each species is Gaussian with a dynamic mean $\bar{z}_i(t)$ and a fixed variance V_i,

$$\phi_i(z,t) = \frac{1}{\sqrt{2\pi V_i}} \exp\left(-\frac{z-\bar{z}_i(t)}{2V_i}\right). \tag{1}$$

2.1. Fecundity and competition

The reproduction and death of an individual are determined by the individual's intrinsic growth function and its mortality due to intraspecific and interspecific competition. Following Pastore et al. (2021), we assume that the intrinsic growth function of individuals with phenotype z in species i is $b_i(z)$, which is defined by a quadratic function,

$$b_i(z) = K_i - gz^2. \tag{2}$$

For simplicity, we use the term "fecundity" to denote the function $b_i(z)$, although this more accurately represents the net growth rate in the absence of competition. The maximum fecundity of an individual in species i is K_i, which is achieved at the optimal trait value $z = 0$, and g determines how quickly fecundity declines when deviating from this optimum. Hence, the competing species differ in the distribution of their relevant quantitative trait and their carrying capacities. Following Pastore et al. (2021), we can interpret $\theta = 1/\sqrt{2g}$ as the width of the fecundity function, with larger values of g reflecting a more narrow function and therefore steeper declines when deviating from the optimum (see Fig. 1). Eq. (2) corresponds to an approximation, for large values of θ, of the Gaussian growth function with variance θ^2 assumed in many classical models of character displacement (Case and Taper, 2000; Day, 2000; Roughgarden, 1976; Lyberger et al., 2025; Taper and Case, 1985), although many of these earlier models assume that the carrying capacity, rather than the fecundity, is a Gaussian function of the trait. Furthermore, using a quadratic function also allows us to avoid the structurally unstable result that infinitely many species can coexist with the Gaussian growth function (Gyllenberg et al., 2005; Pigolotti et al., 2010; Barabas et al., 2012).

As in Pastore et al. (2021), we assume that competition between two individuals with traits z_1 and z_2 is symmetric and given by a Gaussian kernel,

$$a(z_1, z_2) = \exp\left(-\frac{(z_1-z_2)^2}{2\omega^2}\right), \tag{3}$$

where ω is the competition width. Consequently, competition is maximal when individuals have similar trait values ($a(z,z) = 1$), and decreases as the absolute difference between traits z_1 and z_2 increases. All model parameters are summarized in Table 1.

2.2. Eco-evolutionary dynamics

Using these model ingredients, we express the average fecundity of species i, f_i, and competition, α_{ij}, by integrating the fecundity function and competition kernel, respectively, over the trait distributions of the species (Pastore et al., 2021). Integrating $b_i(z)$ with respect to the species' trait weighted by its distribution $\phi_i(z,t)$ defines the average per-capita fecundity of species i in the absence of competition,

$$f_i(\bar{z}_i) = K_i - g\left[(\bar{z}_i)^2 + V_i\right]. \tag{4}$$

Similarly, the average competition between i and j is,

$$\alpha_{ij}(\bar{z}_i, \bar{z}_j) = \frac{\omega}{\Omega_{ij}} \exp\left(-\frac{(\bar{z}_i-\bar{z}_j)^2}{2\Omega_{ij}^2}\right), \tag{5}$$

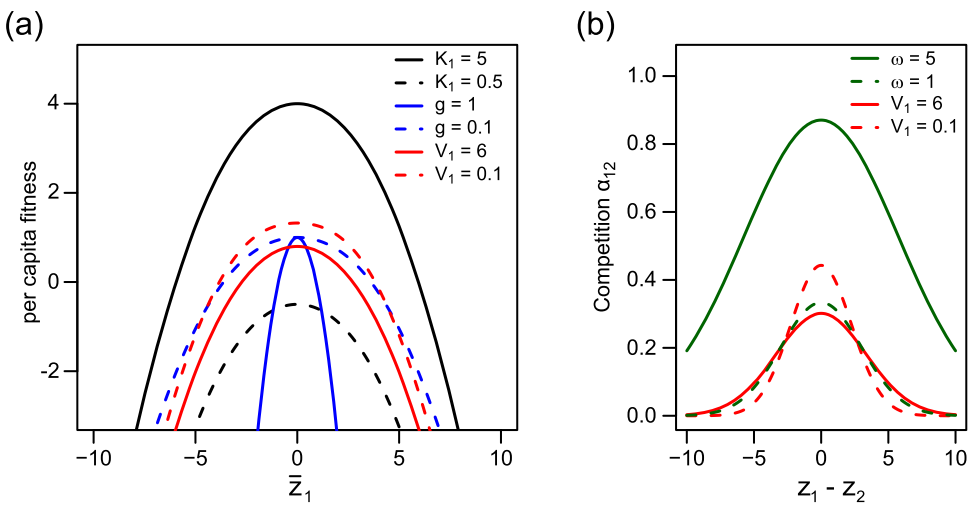

Fig. 1. (a) Effect of high values (solid lines) and low values (dashed lines) of K_1, g, and V_1 on invader per capita fitness as a function of the invader's trait when invader density is 0 and (b) the effect of V_1, and ω on competition as a function of the difference between resident and invader traits. Parameters are as follows unless specified in the panel: $\omega = 1$, g = 0.125, $K_1 = 2$, $K_2 = 1$, $V_1 = V_2 = 4$.

where

$$\Omega_{ij}^2 = \omega^2 + V_i + V_j \tag{6}$$

captures the effective competition width, which depends on the intrinsic competition width ω and on the trait variances of each species. The average intraspecific competition α_{ii} is obtained as the case when $i = j$, leading to $\alpha_{ii} = \omega/\Omega_{ii}$ where $\Omega_{ii}^2 = \omega^2 + 2V_i$ because $\bar{z}^i - \bar{z}^j = 0$ and $V_i = V_j$. The average fecundity (f_i) and competition (α_{ii} and α_{ij}) therefore depend on the trait means and variances of the two species.

Lastly, we calculate the selective effect of interspecific competition by integrating with respect to both traits weighted by each trait distribution $\phi_i(z,t)$ and $\phi_j(z,t)$, which leads to,

$$\beta_{ij}(\bar{z}_i, \bar{z}_j) = -\frac{\omega}{\Omega_{ij}^3}(\bar{z}_i - \bar{z}_j)\exp\left(-\frac{(\bar{z}_i - \bar{z}_j)^2}{2\Omega_{ij}^2}\right). \tag{7}$$

Then, the joint dynamics of the density and mean trait of species i can be written as Pastore et al. (2021),

$$\frac{dn_i}{dt} = n_i\left[f_i - n_i\alpha_{ii} - n_j\alpha_{ij}\right], \qquad i = 1,2 \tag{8}$$

$$\frac{d\bar{z}_i}{dt} = V_i\left[-2g\bar{z}_i - n_j\beta_{ij}\right]. \qquad i = 1,2 \tag{9}$$

Note the term in brackets of Eq. (8) is the per-capita growth rate of species i, and the term in brackets of Eq. (9) is the per-capita growth rate gradient with respect to $\bar{z}_i$ (i.e, the derivative of the term in brackets of Eq. (8) with respect to $\bar{z}_i$). For simplicity, Eq. (9) implicitly assumes that the trait is fully heritable.

We refer the reader to Pastore et al. (2021) for further details on the derivation of these equations. Apart from a few differences in notation and the assumption of fully heritable traits, our model is strictly identical to theirs.

2.3. Model assumptions

The state space of interest for the model (Eqs. 8-9) is $\mathbb{R}_+^2 \times \mathbb{R}^2$. Additionally, our model includes parameters, K_1, K_2, V_1, V_2, ω, and g, which are all assumed to be positive. Finally, we assume that $K_i - gV_i > 0$ for $i = 1,2$. If this condition is reversed, then species i would not persist even in the absence of competition, which is not a scenario of interest.

Solutions of our model equations are well-behaved in the sense that population densities and trait values are (eventually) bounded from above and thus do not diverge. More precisely, we have the following two propositions:

Table 1
Description of parameters in this a two competing species eco-evolutionary model.

Parameter	Description
K_i	maximal fecundity of species i at the optimal trait ($z = 0$)
g	strength of stabilizing selection (decay of fecundity away from optimal trait)
V_i	variance of the trait distribution in species i
ω	competition niche width (competition specificity)

Proposition 1. *Let* $\gamma = \max\left\{\frac{K_1\omega}{\Omega_{11}}, \frac{K_2\omega}{\Omega_{22}}\right\}$. *Then, solutions to (8-9) for any initial condition satisfy*

$$\limsup_{t\to\infty}\left\{n_1(t), n_2(t)\right\} \le \gamma$$

To prove this, first observe that whenever $n_i > \frac{K_i\omega}{\Omega_{ii}}$, the per-capita growth of species i is bounded from above by a negative quantity, i.e.,

$$f_i - n_i\alpha_{ii} - n_j\alpha_{ij} < -gV_i,$$

for $i \neq j$. Therefore, all solutions eventually enter $[0, \frac{K_1\omega}{\Omega_{11}}] \times [0, \frac{K_2\omega}{\Omega_{22}}]$.

Proposition 2. *There is a* z^* *such that solutions to (8-9) for any initial condition, satisfy*

$$\limsup_{t\to\infty}\left\{|z_1(t)|, |z_2(t)|\right\} \le z^*$$

The proof of the second proposition is in the appendix. Altogether, these two propositions give us the desired property of having *dissipative* model equations. That is, there is a compact set $Q \subset \mathbb{R}_+^2 \times \mathbb{R}^2$ that all solutions eventually enter and remain within. In biological terms, the intraspecific competition within each species and the stabilizing selection around zero ensure that the densities and trait values retain biologically realistic values and do not blow up.

3. Resident-invader dynamics

Eqs. (8) and (9) capture the joint dynamics of densities and mean traits for the two competing species. Pastore et al. (2021) used these equations to calculate how the classical metrics of modern coexistence theory (niche overlap and the ratio of competitive differences) change over time. However, the key criterion used to derive these metrics in modern coexistence theory is invasion from rarity. In this section, we first characterise the eco-evolutionary dynamics of the resident (species j) in the absence of the invader (species i). Then, we characterize the

invader trait dynamics according to Eq. (9). In particular, we show that the invader's trait may evolve to either a zero or non-zero equilibrium.

3.1. Equilibrium of the resident species

Proposition 3. *Solutions to equations (8-9) with $n_i(0) = 0$, $n_j(0) > 0$ satisfy,*

$$\lim_{t\to\infty} n_j(t) = \frac{\Omega_{jj}}{\omega}(K_j - gV_j) := \hat{n}_j, \tag{10}$$

$$\lim_{t\to\infty} \bar{z}_j(t) = 0. \tag{11}$$

We give a proof of this proposition in the Appendix. This proposition states that the resident species j will converge to an equilibrium density $\hat{n}_j$ and a mean trait of zero in the absence of a competitor. That is, the mean trait value of species j converges to its optimal trait value without interspecific competition ($\bar{z}_j = 0$). Note that, because $\Omega_{jj}^2 = \omega^2 + 2V_j$, the equilibrium density of species j is non-monotonic with respect to the variance of its trait distribution, V_j. In the Supplementary Fig. 1, we show one example of how the equilibrium population density of the resident depends on its trait variance V_j.

3.2. Eco-evolutionary invasion dynamics

To test for coexistence, community ecologists often examine growth rates of a rare invading species in an environment determined by the resident species. However, when considering eco-evolutionary dynamics, the growth rate of the invading species depends on the distribution of both resident and invader trait values, which are constrained by evolutionary dynamics. In our case, the trait distribution of the invading species i is fully described by the mean trait, $\bar{z}_i$, for a fixed value of the trait variance V_i. Hence, in the context of eco-evolutionary dynamics, we need to ask: for what mean trait values of the resident and invading species should we test invasion? In the following, we show that the mean trait of the invader approaches an equilibrium point, but that this is not necessarily unique, depending on parameter values.

Proposition 4. *Let $h_i(\bar{z}_i) := f_i - \hat{n}_j \alpha_{ij}(\bar{z}_i, 0)$ be the per-capita growth rate of species i at the resident equilibrium of species j given in equations (10-11). Let $n_c = 2g\Omega_{ij}^3/\omega$. Then,*

1. *If $\hat{n}_j < n_c$, then*
   (a) *h_i is unimodal, with a maximum at $\bar{z}_i = 0$.*
   (b) *solutions to equations (8-9) with $n_i(0) = 0$, $n_j(0) > 0$ satisfy $\lim_{t\to\infty} \bar{z}_i(t) = 0$*
2. *If $\hat{n}_j > n_c$, then*
   (a) *h_i is bimodal, with a minimum at $\bar{z}_i = 0$ and maxima at $\hat{\bar{z}}_i^{\pm} := \pm\Omega_{ij}\sqrt{-2\ln\left(\frac{n_c}{\hat{n}_j}\right)}$.*
   (b) *solutions to equations (8-9) with $n_i(0) = 0$, $n_j(0) > 0$ and $\bar{z}_i(0) > 0$ satisfy $\lim_{t\to\infty} \bar{z}_i(t) = \hat{\bar{z}}_i^{+}$*

We give a proof of this proposition in the Appendix. This proposition identifies two different outcomes for the invader's evolutionary dynamics, depending on whether the resident's equilibrium density is above or below a threshold $n_c = 2g\Omega_{ij}^3/\omega$. These two outcomes differ by (1) the shape of the invader's per-capita growth rate at the unique resident equilibrium, in particular the number of peaks (this is characterised by parts 1a and 2a of Proposition 4), and (2) the dynamics of the invader's mean trait, which can converge either to zero or to non-zero equilibrium values. Biologically, this corresponds to a trade-off between maximising intrinsic growth rate (at $\bar{z}_i = 0$) and avoiding interspecific competition, which depends on the population density and trait dynamics of the resident. Hence, the invader's actual growth rate is not necessarily maximised at a trait value of zero.

Which of these conditions is met depends on the strength of stabilising selection towards the fecundity optimum, g, on the competition width, ω, on the trait variances in each species (V_i and V_j), and on the intrinsic fecundity of the resident K_j. When the resident's population density is higher than n_c (e.g. due to a high intrinsic growth rate, K_j, or low trait variance, V_j), condition 2 is met and the per-capita growth functions $h_i(\bar{z}_i)$ are bimodal (Case A, B and C in Fig. 2i). In this case, an invading species with an initial positive mean trait will converge to a fixed positive value $\hat{\bar{z}}_i^{+}$, which is distinct from the optimum of the fecundity function at zero (and by the symmetry of our model, if the invading species has an initial negative mean trait, then this will instead converge to $\hat{\bar{z}}_i^{-}$). Hence, the invading species is expected to differentiate itself from the resident. The invader's per-capita growth rate can be positive at all equilibria (Case A in Fig. 2), positive only at $\bar{z}_i^{\pm}$ but not at the 0 trait equilibrium (Case B), or negative everywhere (Case C).

On the other hand, when the resident's equilibrium density is below the threshold n_c (e.g. due to a low intrinsic fecundity, K_j, or high trait variance, V_j), condition 1 is met and the per-capita growth function $h_i(\bar{z}_i)$ is unimodal (Case D and E in Fig. 2 i). In this case, the invading species with an initial positive mean trait will converge to a value of zero. The per-capita growth can be either positive everywhere (case D) or negative everywhere (case E). Interestingly, the evolution of invading species i does not depend on its maximal fecundity K_i.

Fig. 2(ii) shows how the variances of the invading and resident species lead to each of these 5 invasion cases, and hence, affects the invasion of the rare species. With respect to invasion criteria, we observe that the variance of the resident has two opposing effects. On one hand, it may make invasion (positive per-capita growth of the invader) more likely, since it puts a load on the resident, decreasing its equilibrium population density (see Eq. (10)). On the other hand, it may make invasion less likely since it also decreases intraspecific competition of the resident, thereby potentially increasing its equilibrium population density. In the Supplementary Fig. 2, we show an analogous figure to Fig. 2(ii) for different values of K_1 and K_2.

To shed light on these results, it is interesting to rewrite the threshold density n_c as

$$n_c = \left(\frac{\omega}{\theta}\right)^2 \left(1 + \frac{V_i + V_j}{\omega^2}\right)^{3/2} \tag{12}$$

where $\theta = 1/\sqrt{2g}$ is the width of the fecundity function. Hence, in the absence of trait variances, the threshold n_c corresponds to the square of the ratio of competition width over resource width that characterises whether selection is stabilising or disruptive at the resource optimum in classical character displacement models (Day, 2000; Sasaki and Dieckmann, 2011; Lyberger et al., 2025; Taper and Case, 1985). Non-zero trait variances increase the threshold n_c, but an increase in the invader's variance V_i only affects the denominator of the condition $\hat{n}_j/n_c$ (making it larger), while an increase in the resident's variance V_j affects both numerator and the denominator. In addition, an increase in the resource width θ leads to a lower threhold n_c, which means that the condition $\hat{n}_j > n_c$ is more easily satisfied. High values of θ lead to small values of the ratio ω/θ, which corresponds to the condition for the coexistence of more than one type in Gaussian character displacement models (Roughgarden, 1972; Sasaki and Dieckmann, 2011; Lyberger et al., 2025).

The motivation behind Proposition 4 is similar to the approach proposed by Yamamichi et al. (2022). If we assume that the invader is sufficiently rare that it does not affect the dynamics of the resident population ($n_i \ll n_j$), but sufficiently abundant that we can use a deterministic gradient equation to describe the change of the mean trait in the invader's subpopulation, we can assume that the trait value of the invader will be near its equilibrium value. Then, we can examine invasion at these equilibrium values as a proxy for when the invader is rare. However, unlike in Yamamichi et al. (2022), our analysis shows that the equilibrium trait value at which invasion needs to be evaluated is not necessarily unique. Proposition 4 also extends previous results by Case and Taper (2000) who, using a related model, characterised the resident-invader dynamics and showed that the invader's growth rate could be either unimodal or multimodal depending on a threshold

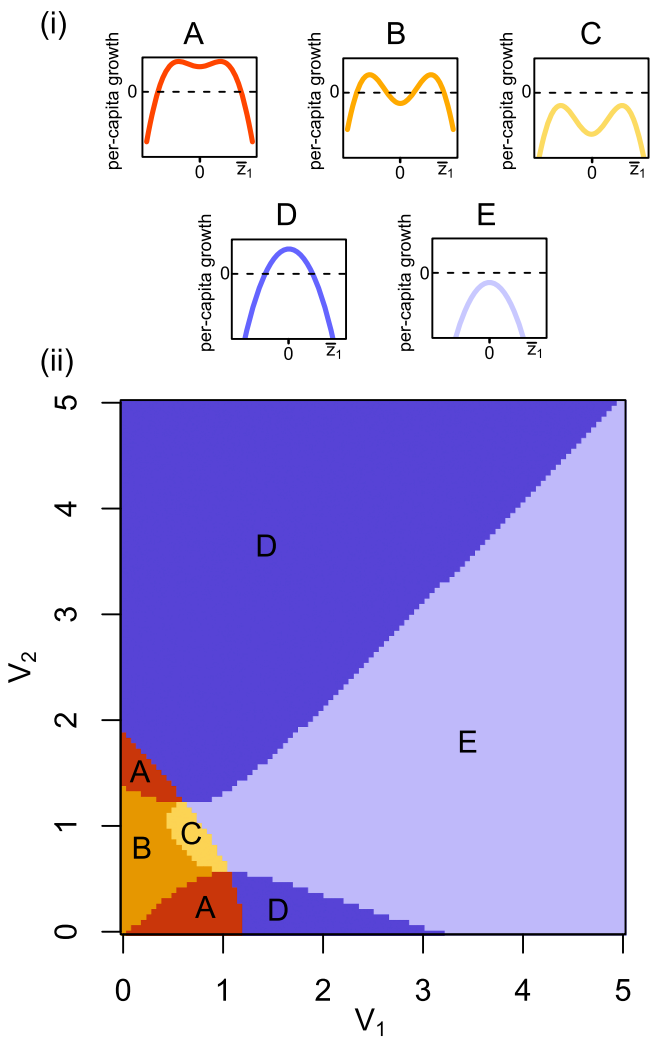

Fig. 2. The top panels (i) display the per-capita growth rate function of an invader i at equilibrium of the resident, $h_i(\bar{z}_i) := f_i - \hat{n}_j \alpha_{ij}(\bar{z}_i, 0)$. In A, B, and C, condition 2 in proposition 4 is met and the function is bimodal, while in D and E condition 1 is met and the function is unimodal. The bottom panel (ii) shows the effect of variance of the invader V_1 and of the resident V_2 on the outcome of a single species invasion. Parameters are as follows unless specified in the panel: $w = 1$, $g = 0.125$, $K_1 = 0.79$ and $K_2 = 0.80$.

condition (their equation (17)), which is similar to our condition $\hat{n}_j > n_c$. Proposition 4 provides a rigorous demonstration and characterisation of the locations of the peaks, and generalises the results of Case and Taper (2000) to more realistic scenarios where the two species do not have equal trait variances.

4. Persistence and nearly persistence theorems

While the idea of invasion when rare as a proxy for coexistence is useful, it does not necessarily guarantee that all species are expected to persist and becomes more nuanced in higher-dimensional dynamics. For example, in three competing species, rock-paper-scissor dynamics, in which a single rare species is able to invade a resident equilibrium, it is possible that not all species coexist (see the recent paper by Ranjan et al. (2024) for more description). Persistence theory offers a rigorous approach to establishing coexistence and ensures that population densities of all species eventually stay above some minimal threshold given that they all initially are present (Schreiber, 2006). In this section, we provide precise conditions for the two species to coexist in the sense of persistence given their dynamics in the model written above.

4.1. Persistence

We begin with the precise definition of persistence:

Definition 1. Model (8-9) is *persistent* if there is a $\beta > 0$ such that for all initial conditions

$$\left(n_1(0), n_2(0), \bar{z}_1(0), \bar{z}_2(0)\right)$$

with $n_1(0) > 0$ and $n_2(0) > 0$, we have that

$$\liminf_{t\to\infty} \left\{n_1(t), n_2(t)\right\} \geq \beta \tag{13}$$

The following result gives sufficient conditions for persistence

Theorem 1. *If for each $i = 1, 2$ and $j = 1, 2$ with $j \neq i$*

$$\frac{\hat{n}_i}{\hat{n}_j} > \frac{\Omega_{ii}}{\Omega_{ij}} \tag{14}$$

or equivalently,

$$\frac{K_i - gV_i}{K_j - gV_j} > \sqrt{\frac{\omega^2 + 2V_j}{\omega^2 + V_i + V_j}}, \tag{15}$$

then model (8-9) is persistent.

We provide a proof of this theorem in the Appendix.

We first observe that when $V_i = V_j$, condition (15) simplifies to $K_i > K_j$ which cannot be simultaneously satisfied by both species. In this case, the species with greater intrinsic fecundity will exclude the other. Differences in variance are therefore a necessary (but not sufficient) condition for persistence. When the two species have equal variances and fecundities ($K_i = K_j$), this is a degenerate case in which the species are identical when their trait distributions are centered at zero.

When $V_i \neq V_j$, we note that there are really two embedded conditions in condition (15) (for $i = 1, j = 2$ and for $i = 2, j = 1$). This can be rewritten as,

$$\sqrt{\frac{\omega^2 + V_1 + V_2}{\omega^2 + 2V_1}} > \frac{K_1 - gV_1}{K_2 - gV_2} > \sqrt{\frac{\omega^2 + 2V_2}{\omega^2 + V_1 + V_2}}. \tag{16}$$

Note that, in this condition, the invasion is evaluated when both $\bar{z}_1 = 0$ and $\bar{z}_2 = 0$. At this value, the resident is at its best trait; it is maximising its intrinsic growth rate. In the case that the invader has a unimodal fitness curve (cf Proposition 4), this is where the intrinsic growth of the invading species is also maximised, despite the increased competition it faces with the resident. In the case that the invader has a bimodal fitness curve, this is the worst case scenario for the "invader"; interspecific competition is maximised and the species will not evolve away from this local minimum (i.e., it is "stuck" at this trait value). This makes it impossible to increase invader per capita growth rate at a trait mean that differs from the resident. Hence, if the invader can invade here, then it can eventually invade starting from any initial mean trait, and it is the only condition that we need to check.

It is interesting to ask what happens if the persistence conditions are not met. The following statement shows such a case with a partial converse statement.

Theorem 2. *If*

$$\frac{\hat{n}_1}{\hat{n}_2} > \frac{\Omega_{11}}{\Omega_{12}}, \quad \textit{and} \tag{17}$$

$$\frac{\hat{n}_2}{\hat{n}_1} < \frac{\Omega_{22}}{\Omega_{12}}, \tag{18}$$

then the system is not persistent and in particular, solutions to equations (8-9) with $n_1(0) > 0$, $\bar{z}_1(0) = 0$ and $\bar{z}_2(0) = 0$ satisfy $\lim_{t\to\infty} n_2(t) = 0$ and $\lim_{t\to\infty} n_1(t) = \hat{n}_1$.

This captures the case when species 1 is able to invade when rare, but species 2 is not able to invade when rare, when both have an initial mean trait of zero. In this case, we see a dynamic analogous to the classical competitive exclusion of species 2.

The proof is fairly straightforward. It relies on recognizing that the set $\{(n_1, n_2, 0, 0) | n_1 \in \mathbb{R}_+, n_2 \in \mathbb{R}_+\}$ is invariant under the model equations. Then, the eco-evolutionary model equations reduce down to the classical Lotka-Volterra ecological equations of two competing species. The conditions given have an equivalence to the classical conditions of species 1 competitively excluding species 2.

4.2. Near persistence

In eco-evolutionary dynamics, persistence, classically defined, can be a somewhat restrictive criteria for thinking about coexistence; it requires that each species can invade for *all* initial trait values. In our model, this includes the zero equilibrium mean trait value of the invader, even when it is a local minimum of a bimodal fitness curve. In this case, the trait value of zero is unstable (in the dynamics restricted to the space where $n_2 = 0$) and small perturbations would allow the invading species to evolve to a different equilibrium value. Hence, it could be the case that the invader is able to invade and subsequently coexist with the resident for some given initial values of its mean trait, i.e. trait-dependent persistence. Because of this, we introduce a less restrictive (but weaker) notion of persistent, called "nearly persistent", that may be more relevant biologically.

Definition 2. Model (8-9) is *nearly persistent* if there is a $\beta > 0$ such that for all initial conditions

$$(n_1(0), n_2(0), \bar{z}_1(0), \bar{z}_2(0))$$

with $n_1(0) > 0, n_2(0) > 0, \bar{z}(0) \neq 0, \bar{z}_2 \neq 0$, we have that,

$$\liminf_{t\to\infty} \{n_1(t), n_2(t)\} \geq \beta.$$

The key distinction is that now we restrict the initial conditions of the trait values to be non-zero. As in persistence, this still ensures that all species will continue to persist despite large perturbations, whether they are ecological (such as a sudden mortality event that disturbs a subset of populations) or evolutionary (such as a sudden selection event).

The next result provides a sufficient condition for our model to be nearly persistent.

Theorem 3. *If, in addition to (17-18), for each $i = 1, 2$ and $j = 1, 2$ with $j \neq i$, we have,*

$$\frac{\hat{n}_i}{n_c} > \frac{\Omega_{ii}}{\Omega_{ij}}\left(1 - \ln \frac{n_c}{\hat{n}_j}\right) \tag{19}$$

or equivalently,

$$K_i - gV_i > 2g\Omega_{ij}^2\left(1 - \ln \frac{2g\Omega_{ij}^3}{\Omega_{jj}(K_j - gV_j)}\right) \tag{20}$$

then model (8-9) is nearly persistent.

We provide a proof of this theorem in the Appendix. Biologically, this statement focuses on the case where each species has a bimodal fitness curve when rare. It only requires invasion of the rare species at the maxima of the bimodal fitness curve when rare, rather than at trait values of 0, i.e., the local minimum of the fitness curve when rare.

Caution is needed to exclude the possibility of eco-evo cycling, in which the population densities of each species get increasingly close to zero even though they eventually invade. This can happen as follows: suppose species 1 is resident, then species 2 invades at its positive trait value and drives species 1 close to zero. When species 1 is rare, species 2 evolves to a trait of zero and species 1 then evolves to a positive trait value, which then leads species 1 to invade. Then, at this trait, species 1 might begin to exclude species 2, and species 2 can get very close to zero density, and the cycle continues. This theorem states that invasion at fitness peaks is indeed sufficient for near persistence and we do not get these types of cycles.

5. Categorization of dynamics and numerical solutions

In this section, we extend beyond persistence to examine the other possible dynamic flows in this model. More specifically, based on pairwise trait-dependent invasion conditions of the invader with a single-resident population, we make predictions on and categorize the two species eco-evolutionary outcomes. These are illustrated in Fig. 3. We then test these conditions numerically. We first distinguish three cases based on the modality of the per-capita growth function of the invader species (i.e., when it has density zero). Then, we partition each case by the possible invasion conditions (i.e, whether growth is positive or negative at the equilibra), which leads to a total of 15 possibilities. The three cases and conditions for the per-capita growth function modality are as follows:

- Case I: If $\hat{n}_1 > n_c$ and $\hat{n}_2 > n_c$.
  In this case, both species have bimodal per-capita growth functions as invaders when the opposing species is a resident at equilibrium. That is, the trait dynamics of the invader has three equilibra.
- Case II: If $\hat{n}_1 < n_c$ and $\hat{n}_2 > n_c$.
  In this case, species 1 has a bimodal per-capita growth function (i.e., three equilibra for its trait dynamics) when species 2 is resident at equilibrium but species 2 has a unimodal per-capita growth function (i.e., unique eq. point at 0), when species 1 is resident at equilibrium.
- Case III: If $\hat{n}_1 < n_c$ and $\hat{n}_2 < n_c$. In this case, both species have unimodal per-capita growth functions as invaders when the opposing species is a resident at equilibrium. That is, the trait dynamics of the invader has a unique equilibrium at 0.

We further partition these cases based on the flow near the boundary. When an invading species has bimodal per capita growth rate curves, we separate the boundary flow into three relevant categories (labeled A, B, C in Fig. 2), while when an invading species has unimodal growth curves, we separate into two relevant categories (labeled D and E in Fig. 2). In (A) and (D), the invading species can invade at all trait equilibria. In (B), the invading species can only invade at the per capita growth peaks. In (C) and (E), the invading species cannot invade at any trait equilibria.

Altogether, the 3 cases and these flows near each boundary result in the 15 possibilities shown in Fig. 3. There are 6 possibilities within Case I, since each species can exhibit one of three invasion categories (A, B, or C). Similarly, there are 6 possibilities within Case II and 3 possibilities within Case III. From numerically examining the 15 possibilities, we observe that there are five qualitatively distinct ecological scenarios possible in this model, including persistence:

1. Persistent: the two competing species will coexist for any initial mean trait values provided both are initially present in the community.
2. Nearly persistent: the two competing species will coexist when either species has non-zero initial mean trait values, provided both are initially present in the community. However, if both species start with an initial trait of zero, then one species will be excluded.
3. Competitive exclusion: species i will competitively exclude species j for any initial trait values. Species j will not evolve to coexist.
4. Near competitive exclusion: species i will competitively exclude species j if either species has a non-zero initial mean trait value. If both species have an initial mean trait value of zero, then there will be a priority effect.
5. Priority effect: for any initial mean trait values of each species, one species will exclude the other, depending on their initial densities.

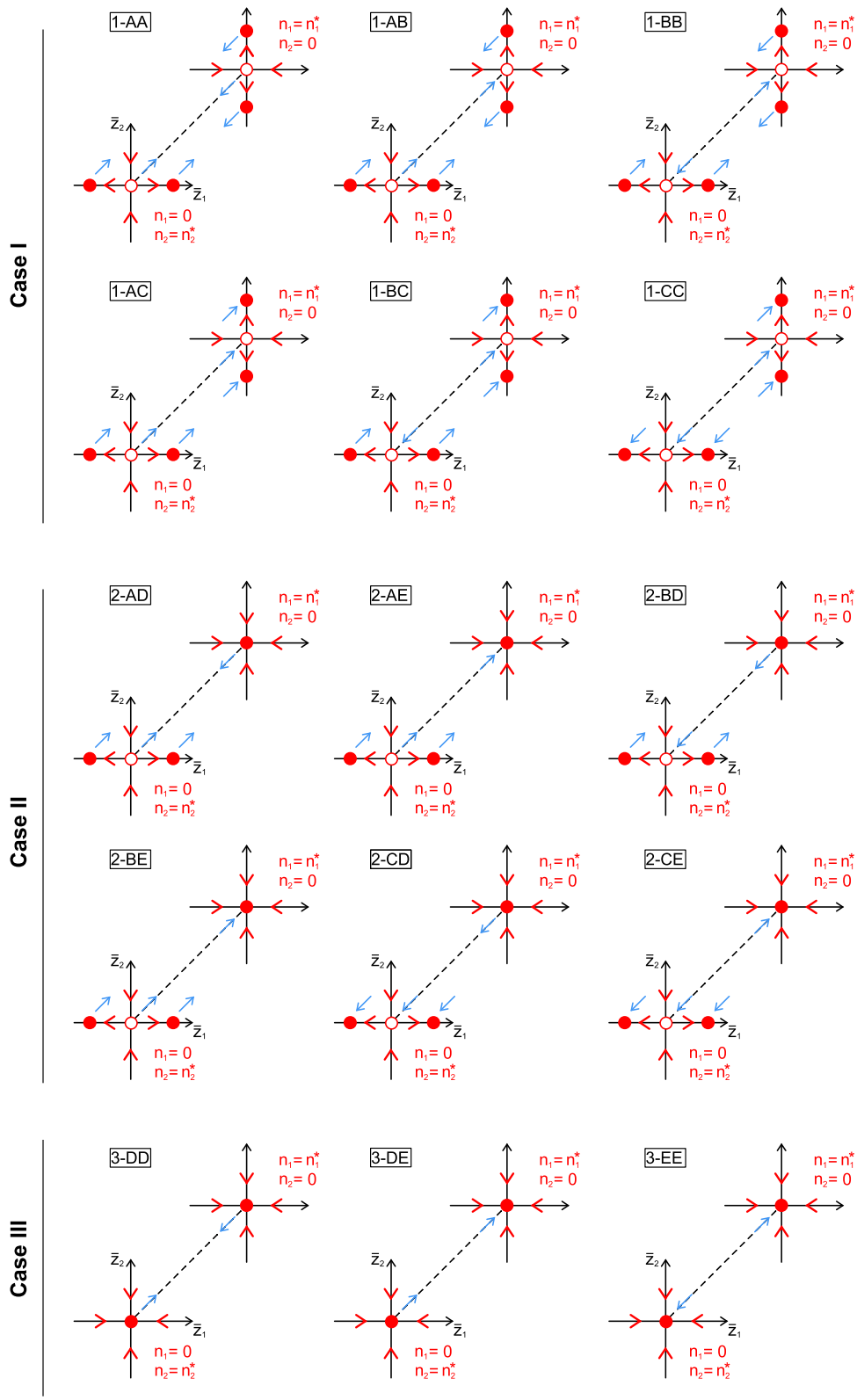

Fig. 3. Two-dimensional representation of four-dimensional space in which the system's dynamics operate. In the front: visualizing the evolutionary dynamics of species 1 is at zero and species 2 is at equilibrium. In the back: visualizing the evolutionary dynamics when species 2 is at zero and species 1 is at equilibrium. Stable and unstable equilibria points (with respect to the trait dynamics on the single-species resident invariant set) are depicted with filled and unfilled circle symbols, respectively. The dotted line indicates the invariant set formed when $\bar{z}_1 = \bar{z}_2 = 0$. See Section 5 for a detailed description of Case I, II and III. Each of these three cases can then be combined with the cases A, B and C described in Fig. 2.

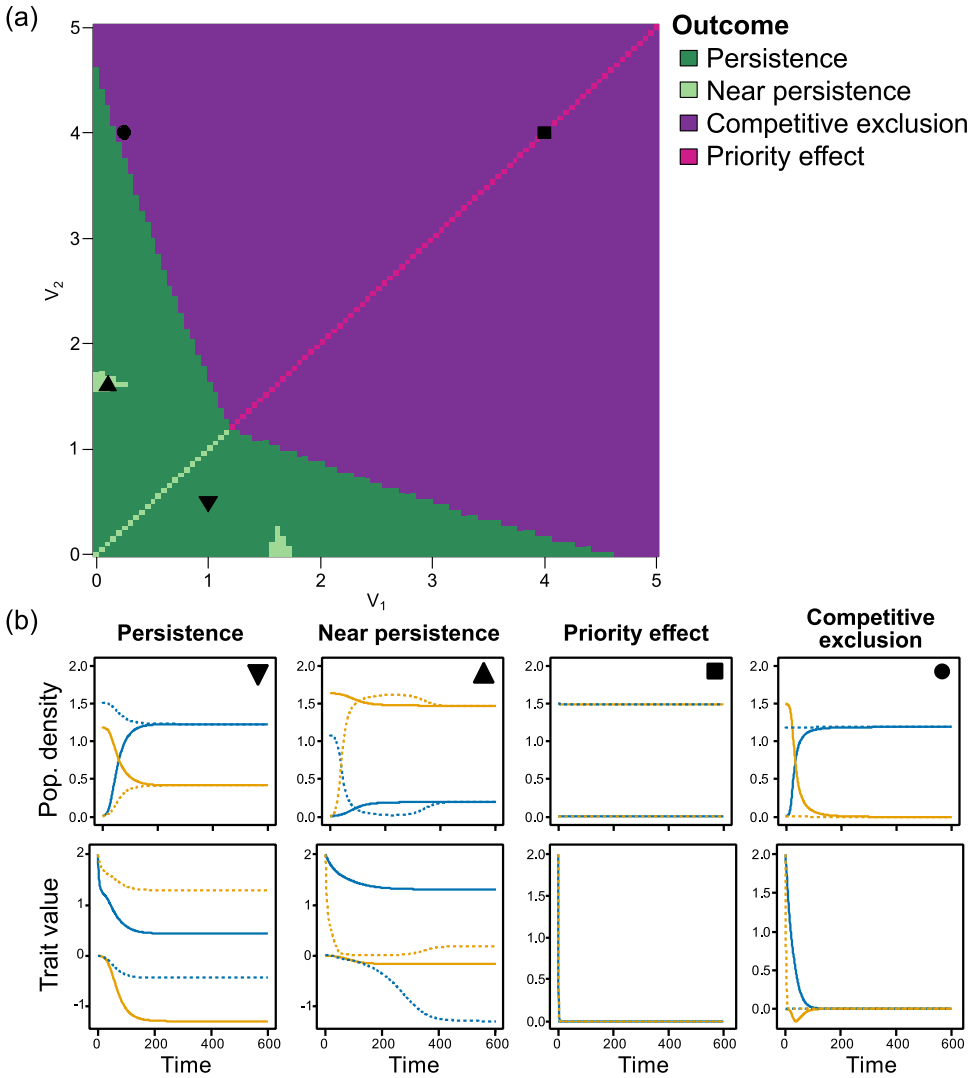

Fig. 4. (a) The effect of changing variance in both species on the qualitative behaviour produced. Parameters are as follows: $w = 1$, $g = 0.125$, $K_1 = K_2 = 1$. The black point represents the values from Pastore et al. 2021. (b) Eco-evolutionary temporal dynamics simulated for three invasion scenarios. Each column shows two stacked panels: the upper panel depicts population densities $n_k(t)$ and the lower panel depicts mean trait values $\bar{z}_k(t)$ for two competing species. Solid lines represent the invasion of species 1 (blue) into a resident population of species 2 (yellow), whereas dashed lines reverse the invasion order. Parameters common to all simulations are competition width $w = 1$, intrinsic growth-rate width $g = 0.125$, and carrying capacities $K_1 = K_2 = 1$. The scenarios differ only in the standard deviations of the Gaussian competition kernels (V_1, V_2): *case 1AA, persistence* (1, 0.25); *case 2DB, nearly persistence* (0.1, 1.6); *case 3DD, priority effect* (4, 4); *case 3DE, competitive exclusion* (0.25, 4). Simulations were run for 600 time units with numerical output every 1 unit. Symbols in panel (b) illustrate simulations with corresponding parameter values in panel (a).

Table 2
Predictions of qualitative behaviors from boundary dynamics and flow near boundary. Letters A-E refer to the per-capita growth rate functions of species 1 and 2 as invading species as shown in Fig. 2. Number in parentheses (i.e., I-III) indicates the boundary dynamic cases (described in the main text). Legend: P = Persistent, NP = Nearly Persistent, CE = Competitive Exclusion, NCE= Near Competitive Exclusion, PE = Priority Effect.

		Species 1				
		A	B	C	D	E
Species 2	A	(I) P	(I) P	(I) CE	(II) P	(II) CE
	B		(I) NP	(I) NCE	(II) NP	(II) NCE
	C			(I) PE	(II) CE	(II) PE
	D				(III) P	(III) CE
	E					(III) PE

Which ecological scenario results in which case of the boundary dynamics and flow is summarized in Table 2. We find that the precise eco-evolutionary outcome we predict strongly depends on the variances and maximal fecundity of the two species. Fig. 4 demonstrates how assuming equal species variances results in near persistence at small variances and priority effects at large variances, whereas a large asymmetry in variances leads to competitive exclusion, where the species with the smaller variance wins. Finally, a small asymmetry in species variances often results in persistence. When maximal fecundity values K differ between species, eco-evolutionary outcomes tend to vary, and we do not observe regions of priority effects. In the Supplementary Fig. 3, we show analogous figures to Fig. 4 for different different values for K_1 and K_2.

6. Discussion

In ecological models, coexistence is typically determined by the criteria of mutual invasion e.g., (Barabás et al., 2018; Chesson, 2000b, 2018; Yamamichi et al., 2022). That is, if each species in the community can invade when its density is low and the rest of the community is at its equilibrium state (e.g., a competitor is at its carrying capacity),

then the species are said to coexist. This is a powerful framework, which indeed has enabled a deeper understanding of the mechanisms underlying coexistence, in modern coexistence theory (Barabás et al., 2018; Chesson, 2000b, 2018). Here, we have expanded on previous attempts to incorporate evolutionary dynamics to understand how trait evolution and intraspecific trait variation affect species coexistence. We find that (1) with eco-evolutionary dynamics, the choice of species traits is not arbitrary when evaluating mutual invasion conditions for an invader and resident species, but determined by the dynamics of the species' mean traits, (2) depending on the trait variances in each species, there can be multiple trait values at which the invasion condition needs to be evaluated to accurately assess the possibility of coexistence, and (3) limitations of the classic invasion criteria (that underly modern coexistence theory) can be overcome by evaluating conditions for persistence. This leads to a more nuanced picture of the role of invasion analyses in determining coexistence.

Our first result is that the evolutionary dynamics of the resident and invader species (e.g. the change in each species' mean traits) determine the trait values at which the classical invasion conditions need to be evaluated. In our 2-species model with quadratic resource function and Gaussian competition kernel, the equilibrium values of the mean traits for the resident and invader can be analytically calculated for a fixed combination of the trait variances. We show that, when the resident density is below a threshold, the invader's trait converges to the optimum trait value for the resident, so that the outcome of the invasion is solely determined by the difference in trait variances in the two species. On the other hand, when the resident density is above a threshold, the invader's trait can converge to either a positive or negative equilibrium value. Overall, this allows us to identify five qualitatively different patterns for invasion, depending on the shape of the per-capita growth rate of invader, which can be either a unimodal or bimodal function of the trait. Our results therefore show that incorporating evolutionary dynamics into coexistence theory is not a straightforward task. Previous studies e.g., (Yamamichi et al., 2022) have emphasised the need to estimate the invasion potential of species at a specific eco-evolutionary (EE) point determined by the coupled eco-evolutionary dynamics. However, the possibility that there could be multiple EE points was not recognised, although earlier character displacement models had hinted at this possibility (Case and Taper, 2000).

The use of invasion criteria as a proxy for coexistence has been particularly successful and widely applied to 2-species communities, and enabled useful interpretations of how the niche overlap and ratio of competitive differences metrics impact coexistence (Barabás et al., 2018; Yamamichi et al., 2022). Persistence theory (Schreiber et al., 2011) is a complementary approach to analysing coexistence, which is more robust in higher-dimensional communities. The second contribution of our study is to apply persistence theory to our eco-evolutionary model to better characterise coexistence of the two species in the sense of persistence. Persistence has a rich history developed in the late 1970s (Freedman and Waltman, 1977; Gard and Hallam, 1979), and early 1980s (Freedman and Waltman, 1984; Gard, 1980, 1981, 1982), but has been limited in the context of eco-evolutionary dynamics (but see Patel and Schreiber, 2018; Browne and Yahia, 2023; Schreiber and Patel, 2015). Hence, applying persistence theory to eco-evolutionary models in the classical notion of persistence requires that population densities eventually remain bounded from below by a positive density *for all* initial trait values. However, in some cases, this might not be the most biologically relevant criterion. Here, we present a weaker notion of persistence: restricting persistence to a subset of initial starting trait values, we can show that the population densities of all species are eventually bounded from below by a positive density. Hence, these communities will continue to persist despite large perturbations to the populations. We anticipate that this notion of coexistence will be useful and generalizable in other contexts, such as eco-evolutionary models with explicit multi-locus genetics (e.g. fixed alleles in one species prevent coexistence while polygenecity preserves it.)

On the technical side, the proof of our near persistent theorem uses previous results from Garay (1989). Roughly speaking, Garay (1989) considered ecological models on a state space in which the boundary of the space is invariant for that model. Typically, the boundary is the extinction set (in which at least one species in a community has density zero). They provided conditions for showing that this set is repelling flow under the model. We recognized that their proof could be applied to other invariant sets and, provided this *includes* the extinction set, we can still show this weaker notion of persistence. Further investigations of persistence in models that include an evolutionary component and how persistence may depend on traits would be worthwhile to build more understanding of how evolution impacts coexistence.

Beyond these technical points, our analysis emphasizes the role of the trait variances of each species in determining coexistence. Some earlier analysis of similar two species models assumed that both species had equal trait variances e.g., (Case and Taper, 2000; Pastore et al., 2021), but this can lead to a degenerate systems when fecundities are also the same. In empirical systems, interacting species will naturally differ in their intraspecific trait variances, with implications for the rate of evolution of each species and for the possibility of species coexistence. Given identical trait variances, Pastore et al. (2021) predicts either near persistence (given small variances) or priority effects (given large variances; Fig. 3a). We observe that when species trait variances differ, species persist or are competitively excluded. In examining the feasability and robustness of coexistence equilibria, Barabás and D'Andrea (2016) conclude that trait variance will generally diminish coexistence, except when trait means are similar and trait variances are sufficiently different. If we think of the trait axis as a resource, our results support the previously established idea that species with similar means and variances will not coexist, due to not being differentiated enough (Barabás and D'Andrea, 2016). However, if both species have similarly small trait variances (i.e., narrow trait distributions), then we predict that species will evolve to differentiate, which can serve as a mechanism of coexistence (niche partitioning). Additionally, if one species has a small trait variance while the other species has a larger trait variance (i.e., broad trait distribution), then species can also coexist, even with similar trait means. In this case, the species with a larger trait variance is able to exploit a broader range of the trait, enabling coexistence via a specialist-generalist distinction (Barabás and D'Andrea, 2016; Bolnick et al., 2011). However, as the trait variances of both species increase, the advantages of differentiation and the possibility of a specialist-generalist form of coexistence are diminished and the two species are not able to coexist.

The other reason why intraspecific trait variation is crucial is that it is central to rapid trait change, and both can affect ecological processes and interactions e.g., (Bolnick et al., 2011; Cope et al., 2022; Kopp and Matuszewski, 2014; Lankau, 2011; Thompson, 1998). The invasion analysis we develop, as the one proposed by Yamamichi et al. (2022), relies on an implicit assumption that the density of the invader is much smaller than the trait variance, so that trait evolution occurs on a fast time scale while the invader is still rare. This is the opposite of the classical assumption of modern coexistence theory, which assumes trait variance to be so small that evolution is negligible and species traits are fixed. A careful analysis of what happens in between these two extreme scenarios is still needed. In addition, species trait variances are unlikely to remain constant as selection can also be expected to alter the shape of the trait distributions. If species trait variances could instead evolve, intraspecific variation would likely decrease as a consequence of interspecific competition (Barabás et al., 2022). This might make persistence more likely in our model because we generally find persistence at lower species trait variances (Fig. 4). Evolution of trait variances might therefore expand the range of initial conditions that result in persistence.

We hope that this analysis supports further investigations into how the persistence of species may depend on their traits. In particular, in this model, we assumed the fecundity was symmetric around the optimal trait value (quadratic function). This assumption leads to the

resident evolving to the optimal trait of zero and zero also being a fixed point for the invading species trait dynamics. Asymmetric fitness functions, which are more biologically realistic, will likely lead to different evolutionary behaviors and, hence, potentially even more nuances in trait-dependent persistence. That is, invading species may be able to invade at some trait values but not others, and hence, coexistence may be more critically dependent on the underlying evolutionary dynamics. Additionally, our model assumes trait distributions remain Gaussian even under the impacts of selection. This is a classical assumption in quantitative genetics, supported by the central limit theorem and presumption that there are a large number of loci that have small additive and independent contributions to a trait. However, it is natural to expect that selection may lead to non-Gaussian distributions (e.g. skewed or multimodal distributions), when, for example, the genetic basis of traits is more complex (see (Turelli and Barton, 1990)) or environmental feedbacks cause disruptive selection that can alter the shape of distributions. One way to better capture the dynamics of the full distribution would be to examine the dynamics of the variance under selection, while assuming higher order moments remain constant. Alternatively, we could incorporate the oligodynamic framework as in Lion et al. (2023), Sasaki and Dieckmann (2011). Finally, here we consider arguably the simplest case of two competing species. Many species of interest live in more complex communities involving several species competing as well as other types of interactions. In these cases, it is natural to predict that there may be even more potential evolutionary outcomes dependent on initial traits. Hence, thinking about the notion of invasion when rare in an eco-evolutionary context may require even more care.

CRediT authorship contribution statement

Swati Patel: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Lynn Govaert:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Kelsey Lyberger:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Victor J. Luque:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **A. Bradley Duthie:** Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Sébastien Lion:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Sebastien Lion reports administrative support and travel were provided by French Foundation for Research on Biodiversity. Alexander Bradley Duthie reports administrative support and travel were provided by German Centre for Integrative Biodiversity Research. Lynn Govaert reports financial support was provided by German Science Foundation. Kelsey Lyberger reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix A: Proofs

We will use standard techniques from dynamical systems and, more specifically, persistence theory, to show the results given in the main text. The proofs rely heavily on the main results from Garay (1989) and Patel and Schreiber (2018) on persistence in ecological systems. These require dissipative systems and so we begin with the proof that our model is dissipative.

A.1. Proof of Proposition 2

We will show that the mean trait value for species 1 is eventually bounded, under our model equations. The same proof applies for species 2. The equation for the trait dynamics of a species 1 is

$$\frac{d\bar{z}_1}{dt} = v_1\left(-2g\bar{z}_1 + n_2(t)\frac{\omega}{\Omega_{ij}^3}(\Delta\bar{z})\exp\left(-\frac{(\Delta\bar{z})^2}{2\Omega_{ij}^2}\right)\right)$$

where $\Delta\bar{z} = \bar{z}_1 - \bar{z}_2$

- First, observe that $(\Delta\bar{z})\exp\left(-\frac{(\Delta\bar{z})^2}{2\Omega_{ij}^2}\right)$ is bounded. That is, there is a δ so that for all $\Delta\bar{z}$, we have
$$|(\Delta\bar{z})\exp\left(-\frac{(\Delta\bar{z})^2}{2\Omega_{ij}^2}\right)| < \delta$$
.
- By proposition 1 eventually in time n_2 is bounded by γ. Hence, there is a T such that for all $t > T$, we have $|n_2(t)\frac{\omega}{\Omega_{ij}^3}(\Delta\bar{z})\exp\left(-\frac{(\Delta\bar{z})^2}{2\Omega_{ij}^2}\right)| < \gamma\frac{\omega}{\Omega_{ij}^3}\delta := \tilde{\gamma}$
- Finally, define $z_1^* := \frac{\tilde{\gamma}}{2g}$. Then, there is a $\beta > 0$ such that for all $t > T$ and $\bar{z}_1 > z_1^*$, we have $\frac{d\bar{z}_1}{dt} < -\beta$ (and if $\bar{z}_1 < -z_1^*$, then $\frac{d\bar{z}_1}{dt} > \beta$). This implies there is a $\tilde{T} > T$ such that for all $t > \tilde{T}$, $z_1(t) < z_1^*$.

We can apply the same procedure to yield a z_2^*. Then let $z^* = \max\{z_1^*, z_2^*\}$.

A.2. Terminology

For the remaining proofs, we will first set up some necessary terminology. As in Patel and Schreiber (2018), our eco-evolutionary model equations can be expressed through the general ecological system with feedback equations:

$$\frac{dn_i}{dt} = n_i h_i(n, z) \qquad i = 1...N \tag{A.1}$$

$$\frac{dz_j}{dt} = g_j(n, z) \qquad j = 1...m \tag{A.2}$$

where n_i is the population density of species i and z_j is a feedback variable.

Let $S \subseteq \mathbb{R}_+^N \times \mathbb{R}^m$ be the state space and $S_0 := \{(n, z) \in S | \prod_{i=1}^N n_i = 0\}$ be the extinction set of this space. Assuming existence and uniqueness of solutions, let $\Phi : \mathbb{R} \times S \to S$ be the flow defined by A.1-A.2. The flow represents the map of the initial condition in the state space to where it is in the state space at some time point later. Also, $\mathcal{O}(y) = \{\Phi(t, y) | t \geq 0\}$ is the forward orbit for initial condition $y = (n, z) \in S$. Assuming dissipativity, there is a compact set $Q \subset S$ so the forward orbits for any initial condition $y \in S$ eventually enter and stay in Q.

The ω-limit set of $y \in S$ is $\omega(y) := \cap_{t\geq 0}\overline{\{\Phi(s,y)|s \geq t\}}$, where the overline indicates the closure of the set (i.e., the set and all its limit points), and the α-limit set of $y \in S$ is $\alpha(y) := \cap_{t\leq 0}\overline{\{\Phi(s,y)|s \leq t\}}$. [Note that here ω-limit is used for identifying a set, which is distinct from the parameter ω.] The ω-limit set of a set $Y \subset S$ is $\omega(Y) := \cap_{t\geq 0}\overline{\{\Phi(s,y)|s \geq t, y \in Y\}}$. A set $Y \subset S$ is invariant if $\{\Phi(t,y)|t \in \mathbb{R}, y \in Y\} = Y$. The global attractor is the maximal compact invariant set $\omega(Q)$.

For an invariant set Γ, we say that $z \in \Gamma$ is chain accessible from $y \in \Gamma$ if for every $\epsilon > 0$, $t > 0$, there is an (ϵ, t)-chain from z to y, i.e., a sequence $\{y_1 = y, y_2, ..., y_{n+1} = z; t_1, t_2, ..., t_n\}$ with $y_i \in \Gamma, t_i > t$ and $d(\Phi(t_i, y_i), y_{i+1}) < \epsilon$ for $i = 1, ..., n$. [Here, d denotes distance between two points.] A point $y \in \Gamma$ is called chain recurrent if it is chain accessible to itself. We let $R_C(\Gamma)$ be the chain recurrent set. Finally, the set Γ is chain recurrent if every $y \in \Gamma$ is chain recurrent.

A collection of sets $\mathcal{M} = \{M_1, M_2, ..., M_\ell\}$ is a Morse decomposition for a compact invariant set Γ if the sets are pairwise disjoint, isolated, invariant, compact sets, so that for every $y \in \Gamma \setminus \cup_{k=1}^{\ell} M_k$ there are integers $i(y)$ and $j(y)$ with $i < j$ and $\omega(y) \subset M_i$ and $\alpha(y) \subset M_j$.

A.3. Proof of Proposition 3-4

Without loss of generality, we will set $i = 1$ (invader) and $j = 2$ (resident). Under the hypothesis $n_1 = 0$, the model equations become

$$\frac{dn_1}{dt} = 0 \tag{A.3a}$$

$$\frac{dn_2}{dt} = n_2(f_2(\bar{z}_2) - n_2\alpha_{22}) \tag{A.3b}$$

$$\frac{d\bar{z}_1}{dt} = v_1(-2g\bar{z}_1 - n_2\beta_{12}(\bar{z}_1, \bar{z}_2)) \tag{A.3c}$$

$$\frac{d\bar{z}_2}{dt} = v_2(-2g\bar{z}_2) \tag{A.3d}$$

The proof relies on recognizing that these are hierarchically decoupled: the flow of $\bar{z}_2$ only depends on $\bar{z}_2$, the flow of n_2 depends on $\bar{z}_2$ and n_2 and the flow of $\bar{z}_1$ depends on n_2, $\bar{z}_2$, and $\bar{z}_1$. We will provide proofs for each claim made in the propositions

- Eq. (A.3d) is a simple linear differential equation with exponential decay solutions $\bar{z}_2(t) = \bar{z}_2(0)\exp(-2gv_2t)$. Hence, $\lim_{t\to\infty} \bar{z}_2(t) = 0$.
- Using the solution of Eq. (A.3d), observe that Eq. (A.3b) becomes an asymptotically autonomous differential equation

$$\frac{dn_2}{dt} = n_2h_1(n_2, t) \tag{A.4}$$

where

$$h_1(n_2, t) = f_2(\bar{z}_2(t)) - n_2\alpha_{22} \to f_2(0) - n_2\alpha_{22} = K_2 - gv_2 - n_2\frac{\omega}{\Omega_{22}}$$

where the convergence is uniform in $\mathbb{R}$. We can make statements about the asymptotically autonomous flow in relation the the autonomous flow, which in this case is

$$\frac{d\tilde{n}_2}{dt} = K_2 - gv_2 - \tilde{n}_2\frac{\omega}{\Omega_{22}} \tag{A.5}$$

The chain recurrent set of (A.5) is the set $\{0, \hat{n}_2\}$, where $\hat{n}_2 = \frac{\Omega_{22}}{\omega}(K_2 - gv_2)$. Then, Theorem 1.8 of Mischaikow et al (1995) gives us that the ω-limit of $n_2(0)$ for (A.4) is contained in $\{0, \hat{n}_2\}$. Finally, if $n_2(0) > 0$ then there is an $\epsilon_1 > 0$ and $T_1 > 0$ so that $n_2(t) > \epsilon_1$ for all $t > T_1$. To see this, observe that by the assumption $K_2 > gv_2$, we have that there is a $T_2 > 0$ and $\epsilon_2 > 0$ so that $h_1 > \epsilon_2$ for all $n_2 < \epsilon_1$ and $t > T_2$. Hence, if $n_2(0) > 0$ then $\lim_{t\to\infty} n_2(t) = \frac{\Omega_{22}}{\omega}(K_2 - gv_2)$.
- In a similar fashion, Eq. (A.3c) is

$$\frac{d\bar{z}_1}{dt} = h_2(\bar{z}_1, t)$$

where

$$h_2(\bar{z}_1, t) = v_1(-2g\bar{z}_1 - n_2(t)\beta_{12}(\bar{z}_1, \bar{z}_2(t)) \to v_1(-2g\bar{z}_1 - \hat{n}_2\beta_{12}(\bar{z}_1, 0))$$

$$= v_1(-2g\bar{z}_1 - \hat{n}_2\frac{\omega}{\Omega_{12}^3}\exp(-\frac{\bar{z}_1^2}{2\Omega_{12}^2}))$$

and again this convergence is uniform in $\mathbb{R}$. We define the autonomous differential equation

$$\frac{d\tilde{\bar{z}}_1}{dt} = v_1(-2g\tilde{\bar{z}}_1 - \hat{n}_2\frac{\omega}{\Omega_{12}^3}\exp(-\frac{\tilde{\bar{z}}_1^2}{2\Omega_{12}^2})) \tag{A.6}$$

The flow defined by (A.6) has two possibilities for its chain recurrent set:

(a) if $K_2 - gv_2 < \frac{2g\Omega_{12}^3}{\Omega_{22}}$, then the chain recurrent set is $\{0\}$

(b) if $K_2 - gv_2 > \frac{2g\Omega_{12}^3}{\Omega_{22}}$, then the chain recurrent set is $\{0, \pm\hat{\bar{z}}_1\}$

By a similar argument to above, we can show that in case 2 and $\bar{z}_1(0) > 0$, the non-autonomous flow defined by (A.3c) has $\lim_{t\to\infty} \bar{z}_i(t) = \hat{\bar{z}}_i$

A.4. Proof of Theorem 1

The statement is:
If for each $i = 1, 2$ and $j = 1, 2$ with $j \neq i$

$$\frac{K_i - gV_i}{K_j - gV_j} > \sqrt{\frac{\omega^2 + 2V_j}{\omega^2 + V_i + V_j}} \tag{A.7}$$

then our model is permanent.

We will use the following theorem (from Patel and Schreiber, 2018) to prove this statement.

Theorem 4. *Let $\mathcal{M} = \{M_1, M_2, ..., M_\ell\}$ be a Morse decomposition for $S_0 \cap \Gamma$ where Γ is the global attractor for A.1. If, for each $M_k \in \mathcal{M}$, there exists a $p_{k1}, p_{k2}, ..., p_{kn} > 0$ such that for every $y \in M_k$, there is a T_y such that*

$$\sum_{i=1}^{N} p_{ki}\int_0^{T_y} h_i(\phi(t,y))dt > 0$$

then A.1 is persistent.

This theorem extends persistence theory results of purely ecological systems to account for feedbacks such as the evolutionary one we consider in this paper. In the model considered in the present work,

$$h_i(n_1, n_2, \bar{z}_1, \bar{z}_2) = f_i - n_i\alpha_{ii} - n_j\alpha_{ij}$$

The proof proceeds in two parts. First, we identify an appropriate Morse decomposition for our system. The Morse decomposition depends on which case we consider. Then, we will show, for each Morse set, the existence of a $\vec{p_k} > 0$ to satisfy the inequality in Theorem 4.

- For Case I, we define the Morse decomposition as the following set of ordered equilibrium points: $M_7 = (0,0,0,0)$, $M_6 = (\hat{n}_1, 0, 0, 0)$, $M_5 = (0, \hat{n}_2, 0, 0)$, $M_4 = (\hat{n}_1, 0, 0, \hat{\bar{z}}_2)$, $M_3 = (0, \hat{n}_2, \hat{\bar{z}}_1, 0)$, $M_2 = (\hat{n}_1, 0, 0, -\hat{\bar{z}}_2)$, $M_1 = (0, \hat{n}_2, -\hat{\bar{z}}_1, 0)$. (These are partially illustrated in Fig. 3.)
- Next, observe that the inequality in Theorem 4 for two species is

$$p_{k1}\int_0^{T_y} h_1(\Phi(t,y))dt + p_{k2}\int_0^{T_y} h_2(\Phi(t,y))dt > 0$$

For each set M_k and $z \in M_k$, we let $\vec{p_k} = (1,1)$ and $T_y = 1$. We show this inequality is met. (In fact, any positive vector and any $T_y > 0$ would do.) Then, recognizing that each Morse set is an isolated equilibrium point, the condition, for Morse set M_k becomes

$$h_1(M_k) + h_2(M_k) > 0 \tag{A.8}$$

- For M_7, the standing assumptions that $K_i - gv_i > 0$ for $i = 1, 2$ ensure inequality A.8 is met.

- Furthermore, for evenly indexed Morse set, M_k, we have $h_1 = 0$ and hence, need only check

$$h_2(M_k) > 0 \tag{A.9}$$

For oddly indexed Morse sets, except M_7, we have $h_2 = 0$ and need only check

$$h_1(M_k) > 0 \tag{A.10}$$

Hence, this persistent condition reduces to checking invasion (positive growth) of an invader j at Morse sets with purely resident species i.
- For M_6 and M_5, the hypothesis of the theorem ensures that $h_2(M_6) > 0$ and $h_1(M_5) > 0$.
- Next, observe that if $h_2(M_6) > 0$, then $h_2(M_4) > 0$ and $h_2(M_2) > 0$. This is true since $\alpha_{12}(0, z) \leq \alpha_{12}(0, 0)$ for all $z \in \mathbb{R}$.
- An analogous argument holds to show that if $h_1(M_5) > 0$, then $h_1(M_3) > 0$ and $h_1(M_1) > 0$.
- Finally, the proof for these Cases II and Case III follows similarly to Case I.

A.5. Proof of Theorem 3

To show the result on near persistence, we use an earlier persistence result from Garay (1989). Garay used the Ura-Kimura Theorem, which roughly states that if the stable set of a compact isolated invariant set is contained in itself, then it is asymptotically unstable. Here, the stable set for an invariant set is

$$W^+(\Gamma) = \{x \in S| \quad \omega(x) \subset \Gamma\}$$

That is, the stable set is the set of all points whose ω-limit is contained in that invariant set. Garay used this theorem to give persistence conditions by showing the boundary of a locally compact metric space is unstable.

Theorem 5. *[Garay] Let $(\mathcal{E}, d)$ be a locally compact metric space and let S be a closed subset of $\mathcal{E}$ and ϕ a dissipative dynamical system for which the boundary ∂S is invariant (and Γ is a maximal compact invariant set). Further, assume that for each component C of the chain recurrent set $R_C(\Gamma \cap \partial S)$, we have*

1. *a $\gamma > 0$ such that $\{x \in S \backslash \partial S | d(x, C) < \gamma\}$ contains no entire trajectories*
2. *$W^+(C) \cap S \backslash \partial S = \emptyset$*

Then, the system is persistent.

In the proof Garay uses the Ura-Kimura theorem to show the boundary ∂S is asymptotically unstable, by examining a corresponding chain recurrent set, and that this then implies persistence. Going through the proof, we recognize that the result of asymptotic instability applies more generally to any non-open invariant set, not just boundary sets. Hence, we can apply this proof to show the invariant set $H := \{(n_1, n_2, \bar{z}_1, \bar{z}_2) | n_1 n_2 = 0\} \cup \{(n_1, n_2, \bar{z}_1, \bar{z}_2) | \bar{z}_1 = \bar{z}_2 = 0\}$ is asymptotically unstable. Here H is the union of the extinction set, i.e., where at least one of the species is extinct, and the set where both species have mean trait of zero.

To apply this theorem, we focus on the chain recurrent set $R_C(\Gamma \cap H)$. These are the equilibrium points:

- $C_1 := (0, 0, 0, 0)$
- $C_2 := (0, \hat{n}_2, 0, 0)$
- $C_3 := (\hat{n}_1, 0, 0, 0)$
- $C_4 := (0, \hat{n}_2, \hat{\bar{z}}_1^-, 0)$
- $C_5 := (0, \hat{n}_2, \hat{\bar{z}}_1^+, 0)$
- $C_6 := (\hat{n}_1, 0, 0, \hat{\bar{z}}_2^-)$
- $C_7 := (\hat{n}_1, 0, 0, \hat{\bar{z}}_2^+)$

Notice that for each pair of components, C_i and C_j, with $i \neq j$, C_i and C_j are not chain accessible from each other. Next, we show that C_1 is isolated and that $W^+(C_1) \cap S \backslash H = \emptyset$. Suppose that this is not true. We let $B_\epsilon(C_1)$ denote the ϵ–neighborhood around C_1, i.e., all points within an ϵ distance from C_1. Then, there is a $y \in B_\epsilon(C_1) \backslash H$ so that $\phi(y, t) \in B_\epsilon(C_1)$ for all $t \geq 0$. From our standing assumptions, there is a $\delta > 0$ and $\epsilon > 0$ so that $h_1(y) > \delta$ and $h_2(y) > \delta$ for all $y \in B_\epsilon(C_1)$ (h_1 and h_2 are the per-capita growth rates as defined in previous proof). Since $y \in B_\epsilon(C_1) \backslash H$ implies either that $n_1 > 0$ or $n_2 > 0$, then there is a $t > 0$ such that $\phi(y, t) \notin B_\epsilon(C_1)$, which is a contradiction. A similar proof holds for $C_2, ..., C_7$. Hence, conditions 1 and 2 of the theorem are met and H is asymptotically unstable.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.jtbi.2026.112443.

References

Allesina, S., Tang, S., 2012. Stability criteria for complex ecosystems. Nature 483 (7388), 205–208. https://doi.org/10.1038/nature10832

Barabás, G., D'Andrea, R., 2016. The effect of intraspecific variation and heritability on community pattern and robustness. Ecol. Lett. 19 (8), 977–986. https://doi.org/10.1111/ele.12636

Barabás, G., D'Andrea, R., Stump, S.M., 2018. Chesson's coexistence theory. Ecol. Monographs 88 (3), 277–303. https://doi.org/10.1002/ecm.1302

Barabás, G., Parent, C., Kraemer, A., Van de Perre, F., De Laender, F., 2022. The evolution of trait variance creates a tension between species diversity and functional diversity. Nat. Commun. 13 (1), 2521.

Barabás, G., Parent, C., Kraemer, A., de Perre, F.V., De Laender, F., 2022. The evolution of trait variance creates a tension between species diversity and functional diversity. Nat. Commun. 13. https://doi.org/10.1038/s41467-022-30090-4

Barabás, G., Pigolotti, S., Gyllenberg, M., Dieckmann, U., Meszéna, G., 2012. Continuous coexistence or discrete species? A new review of an old question. IR-12-020

Bolnick, D.I., Amarasekare, P., Araújo, M.S., Bürger, R., Levine, J.M., Novak, M., Rudolf, V. H.W., Schreiber, S.J., Urban, M.C., Vasseur, D.A., 2011. Why intraspecific trait variation matters in community ecology. Trends Ecol. Evol. 26, 183–192. https://doi.org/10.1016/j.tree.2011.01.009

Browne, C.J., Yahia, F., 2023. Virus-immune dynamics determined by prey-predator interaction network and epistasis in viral fitness landscape. J. Math. Biol. 86 (1), 9.

Brunner, F.S., Anaya-Rojas, J.M., Matthews, B., Eizaguirre, C., 2017. Experimental evidence that parasites drive eco-evolutionary feedbacks. Proc. Nat. Acad. Sci. 114 (14), 3678–3683.

Case, T.J., Taper, M.L., 2000. Interspecific competition, environmental gradients, gene flow, and the coevolution of species' borders. Am. Nat. 155 (5), 583–605.

Chesson, P., 2018. Updates on mechanisms of maintenance of species diversity. J. Ecol. 106, 1773–1794. https://doi.org/10.1111/1365-2745.13035

Chesson, P.L., 1982. The stabilizing effect of a random environment. J. Math. Biol. 15, 1–36.

Chesson, P.L., 2000a. General theory of competitive coexistence in spatially-varying environments. Theor. Popul. Biol. 237, 211–237.

Chesson, P.L., 2000b. Mechanisms of maintenance of species diversity. Ann. Rev. Ecol. Syst. 31, 343–366.

Chesson, P.L., Warner, R.R. R.R., 1981. Environmental variability promotes coexistence in lottery competitive systems. Am. Nat. 117 (6), 923–943.

Cope, O.L., Burkle, L.A., Croy, J.R., Mooney, K.A., Yang, L.H., Wetzel, W.C., 2022. The role of timing in intraspecific trait ecology. https://doi.org/10.1016/j.tree.2022.07.003

Cropp, R., Norbury, J., 2021. The eco-evolutionary modelling of populations and their traits using a measure of trait differentiation. J. Theor. Biol. 531, 110893.

Day, T., 2000. Competition and the effect of spatial resource heterogeneity on evolutionary diversification. Am. Nat. 155 (6), 790–803. https://doi.org/10.1086/303356

Dougoud, M., Vinckenbosch, L., Rohr, R., Bersier, L.-F., Mazza, C., et al., 2018. The feasibility of equilibria in large ecosystems: a primary but neglected concept in the complexity-stability debate. PLOS Comput. Biol. 14 (2), e1005988. http://arxiv.org/abs/1612.06735.

Ellner, S.P., Geber, M.A., Hairston, N.G., 2011. Does rapid evolution matter? Measuring the rate of contemporary evolution and its impacts on ecological dynamics. Ecol. Lett. 14 (6), 603–614. https://doi.org/10.1111/j.1461-0248.2011.01616.x

Farkas, T.E., Mononen, T., Comeault, A.A., Hanski, I., Nosil, P., 2013. Evolution of camouflage drives rapid ecological change in an insect community. Current Biol. 23 (19), 1835–1843.

Freedman, H.I., Waltman, P., 1977. Mathematical analysis of some three-species food-chain models. Math. Biosci. 33 (3-4), 257–276.

Freedman, H.I., Waltman, P., 1984. Persistence in models of three interacting predator-prey populations. Math. biosci. 68 (2), 213–231.

Garay, B.M., 1989. Uniform persistence and chain recurrence. J. Math. Ana. Appl. 139 (2), 372–381.

Gard, T.C., 1980. Persistence in food webs: Holling-type food chains. Math. Biosci. 49 (1-2), 61–67.

Gard, T.C., 1981. Persistence for ecosystem microcosm models. Ecol. Modell. 12 (4), 221–229.

Gard, T.C., 1982. Top predator persistence in differential equation models of food chains: the effects of omnivory and external forcing of lower trophic levels. J. Math. Biol. 14 (3), 285–299.

Gard, T.C., Hallam, T.G., 1979. Persistence in food webs–i lotka-volterra food chains. Bullet. Math. Biol. 41 (6), 877–891.

Govaert, L., Fronhofer, E.A., Lion, S., Eizaguirre, C., Bonte, D., Egas, M., Hendry, A.P., De Brito Martins, A., Melián, C.J., Raeymaekers, J. A.M., Ratikainen, I.I., Saether, B.E., Schweitzer, J.A., Matthews, B., 2019. Eco-evolutionary feedbacks–Theoretical models and perspectives. Funct.Ecol. 33 (1), 13–30. arXiv preprint arXiv:1806.07633. https://doi.org/10.1111/1365-2435.13241

Grilli, J., Adorisio, M., Suweis, S., Barabás, G., Banavar, J.R., Allesina, S., Maritan, A., et al., 2017. Feasibility and coexistence of large ecological communities. Nat. Commun. 8. https://doi.org/10.1038/ncomms14389

Gyllenberg, M., Meszéna, G., 2005. On the impossibility of coexistence of infinitely many strategies. J. Math. Biol. 50(2), 133–160.

Hairston Jr N. G., Ellner S. P., Geber M. A. , Yoshida T. , and Fox J. A., 2005, Rapid evolution and the convergence of ecological and evolutionary time, Ecol. Lett., 8 (10) 1114-1127, https://doi.org/10.1111/j.1461-0248.2005.00812.x.

Hendry A. P., 2016 Eco-evolutionary dynamics, Princeton University Press, Princeton, NJ.

Hanski, I.A., Ranta, E., 1983. Coexistence in a patchy environment: three species of Daphnia in rock pools. J. Animal Ecol. 52 (1), 263–279. http://www.jstor.org/stable/10.2307/4599.

Kopp, M., Matuszewski, S., 2014. Rapid evolution of quantitative traits: Theoretical perspectives. Evol. Appl. 7, 169–191. https://doi.org/10.1111/eva.12127

Lankau, R.A., 2011. Rapid evolutionary change and the coexistence of species. Ann. Rev. Ecol. Evol. Syst. 42, 335–354. https://doi.org/10.1146/annurev-ecolsys-102710-145100

Li, L., Chesson, P.L., 2016. The effects of dynamical rates on species coexistence in a variable environment: the paradox of the plankton revisited. Am. Nat. 188 (2), E000. https://doi.org/10.1086/687111

Lion, S., 2018. Theoretical approaches in evolutionary ecology: environmental feedback as a unifying perspective. Am. Nat. 191 (1). https://doi.org/10.1086/694865

Lion, S., Sasaki, A., Boots, M., 2023. Extending eco-evolutionary theory with oligomorphic dynamics n/a (n/a). https://doi.org/10.1111/ele.14183

Lyberger K., Govaert L., Duthie B., Patel S., Luque V., and Lion S., 2025.

MacArthur, R.H., Wilson, E.O., 1967. The Theory of Island Biogeography. Princeton University Press, Princeton, N.J.

May, R.M., 1972. Will a large complex system be stable? Nature 238, 413–414.

Nee, S., May, R.M., 1992. Dynamics of metapopulations: habitat destruction and competitive coexistence. J. Animal Ecol. 61 (1), 37–40. http://www.jstor.org/stable/10.2307/5506.

Pastore, A.I., Barabás, G., Bimler, M.D., Mayfield, M.M., Miller, T.E., 2021. The evolution of niche overlap and competitive differences 5, 330–337. https://doi.org/10.1038/s41559-020-01383-y

Patel, S., Cortez, M.H., Schreiber, S.J., 2018. Partitioning the effects of eco-evolutionary feedbacks on community stability. Am. Nat. 191 (3), 1–29. https://doi.org/10.1101/104505

Patel, S., Schreiber, S.J., 2018. Robust permanence for ecological equations with internal and external feedbacks. J. Math. Biol. 77, 79–105. https://doi.org/10.1007/s00285-017-1187-5

Pigolotti, S., López, C., Hernández-Garcla, E., Andersen, K.H., 2010. How Gaussian competition leads to lumpy or uniform species distributions. Theor. Ecol. 3 (2), 89–96.

Ranjan, R., Koffel, T., Klausmeier, C.A., 2024. The three-species problem: Incorporating competitive asymmetry and intransitivity in modern coexistence theory. Ecol. Lett. 27 (4), 1–12. https://doi.org/10.1111/ele.14426

Roughgarden, J., 1972. Evolution of niche width. Am. Nat. 106, 683–718.

Roughgarden, J., 1976. Resource partitioning among competing species–a coevolutionary approach. Theor. Pop. Biol. 9 (3), 388–424. https://doi.org/10.1016/0040-5809(76)90054-X

Schreiber, S.J., 2006. Persistence despite perturbations for interacting populations. J. Theor. Biol. 242 (4), 844–852. https://doi.org/10.1016/j.jtbi.2006.04.024

Sasaki A. and Dieckmann U., 2011 Oligomorphic dynamics for analyzing the quantitative genetics of adaptive speciation, J. math. Biol., 63 (4), 601-635, https://doi.org/10.1007/s00285-010-0380-6.

Schreiber, S.J., Benaïm, M., Atchadé, K. A.S., 2011. Persistence in fluctuating environments. J. Math. Biol. 62, 655–683. https://doi.org/10.1007/s00285-010-0349-5

Schreiber, S.J., Patel, S., 2015. Evolutionary induced alternative states and coexistence in systems with apparent competition. Nat. Res. Model. 28 (4), 475–496. https://doi.org/10.1111/nrm.12076

Soetaert, K., Petzoldt, T., Setzer, R.W., 2010. Solving differential equations in r: package desolve. J. Stat. Softw. 33, 1–25.

Taper M. L. and Case T. J., (1985) Quantitative Genetic Models for the Coevolution of Character Displacement, Ecology, 66 (2), 355-371.

Thompson, J.N., 1998. Rapid evolution as an ecological process. Trends . Ecol. Evol. 13 (8), 329–332.

Turelli M. and Barton N. H., Dynamics of polygenic characters under selection, Theor. Pop. Biol., 38 (1) (1990), 1-57, https://doi.org/10.1016/0040-5809(90)90002-D.

Vellend, M., 2016. The theory of ecological communities. Princeton University Press, Princeton. 1515 (97814), 00883.

Weyerer, F., Weinbach, A., Zarfl, C., Allhoff, K.T., 2023. Eco-evolutionary dynamics in two-species mutualistic systems: one-sided population decline triggers joint interaction disinvestment. Evol. Ecol. 37 (6), 981–999.

Yamamichi, M., Gibbs, T., Levine, J.M., 2022. Integr. Eco-Evol. Dyn. Modern Coexistence Theory 25 (10), 2091–2106. https://doi.org/10.1111/ele.14078

Zamorano, L.S., Gompert, Z., Fronhofer, E.A., Feder, J.L., Nosil, P., 2023. A stabilizing eco-evolutionary feedback loop in the wild. Current Biol. 33 (15), 3272–3278.